2022 4th International Conference on Frontiers Technology of Information and Computer (ICFTIC) | 979-8-3503-2195-1/22/$31.00 ©2022 IEEE | DOI: 10.1109/ICFTIC57696.2022.10075212

A New Image Recognition Combining Transfer Learning Algorithm and MobileNet V2 Model for Palm Vein Recognition

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Abstract. **Among human biometrics, palm vein recognition is characterised by high security and privacy as an internal human biometric identification technology. Despite the breakthroughs associated with palm vein technology, there are various factors such as lighting variations, ambient temperature, and angular hand position that make it difficult to extract palm vein features from the original image. Also, due to the lack of sufficient palm vein datasets, the models will not have enough samples to differentiate the features and will easily lead to overfitting of the data. To address the above issues, we propose a MobileNet V2 convolutional neural network model combining a transfer learning algorithm named SEM. It consists of proposing a region of interest (ROI) segmentation method that contains more vein information, and transferring the pre-trained model weights from the palm print dataset to the palm vein recognition domain. Furthermore, the SENet module is embedded into the MobileNet V2 network to enhance the perceptual field of the network, and the important features are reinforced to improve the accuracy. Finally, the proposed SEM model is trained on the dataset, and our proposed method has a very high accuracy and high robustness, with an accuracy of 99.35%.**

Keywords: Deep learning, Palm vein recognition, ROI Segmentation, Transfer-learning, MobileNet V2

I. Introduction

With the progress of society and the rapid development of artificial intelligence, the protection of personal privacy and information security has become a top priority. Biometric recognition is a technology that uses the human body's own biometric features for identity authentication. At present, face recognition, iris recognition, fingerprint recognition, palm vein recognition and other technologies have been widely used in government, banking, security and defense fields. Although human biometric identification technology has brought us great convenience, there are still some potential problems with human biometric identification. For example, face recognition is subject to various occlusions, angles and ambient light sources. Iris recognition is expensive and infrared cameras can cause discomfort. Fingerprint wear and dirt over time can lead to identification failure, and fingerprint traces can be used to replicate fingerprints [1]. Palm vein recognition is a major advantage in the light of the COVID-19 epidemic [2]. In addition, palm vein patterns are more complex and have more unique characteristics than fingerprints or finger vein patterns. The palm vein is an internal biometric feature hidden under the skin, which is stable, private and can only be captured from a living body by near infrared light and is not easily replicated or forged. As a result, palm vein recognition offers a higher level of privacy and security. Recently, with the breakthrough of deep learning in artificial intelligence, many scholars have used transfer learning to solve the problem of insufficient datasets and achieved good results [3]. In this paper, we propose a new image recognition combining transfer learning algorithm and MobileNet V2 model for palm vein recognition.The steps are as follows: First, to minimise the effect of intra-object variation, we pre-process the captured palm vein image and segment a fixed region of interest (ROI) from it. We use transfer learning to address the problem of insufficient training datasets for palm vein recognition. The pre-processed images are then sent to the convolutional neural network for training, while the SENet module is embedded in the MobileNet V2 network, which has the feature of automatically learning the importance of each feature channel and then increasing the weight of important features based on this feature. The motivation and contributions of this research can be summarised as follows:

1) A new region of interest (ROI) segmentation technique is proposed, which achieves a larger ROI regardless of the rotation angle in the horizontal plane relative to the scanner in terms of recognition quality.

2) Transfer learning is used to address the problem that insufficient palm vein datasets may lead to overfitting of the model, thus ensuring the accuracy and stability of the model with stronger generalization capability.

3) We embed the SENet module into a convolutional neural network so that the model automatically learns the importance of different feature channels, reducing the error due to background noise and improving the accuracy of the model.This proposed SEM is tested on a palm vein dataset and its recognition accuracy is compared with other methods, achieving impressive recognition results.

II. Related Work

Vein recognition is an emerging infrared biometrics technology, it is based on the characteristics of venous blood deoxygenated haematoxylin absorbing near infrared rays or far infrared rays radiated by the human body, using the corresponding wavelength range of infrared cameras to capture the vein information of the palm, fingers, back of the hand or wrist, and then extract the vein features through deep learning methods, or normalisation, denoising and other pre-processing for image enhancement, and then extract its features, and match it with the pre-registered database to determine personal identity. The features are then extracted and matched with the vein information pre-registered to the database to determine the identity of the individual. Vein recognition uniquely identifies a person because of the uniqueness of each person's vein information and its permanence after adulthood. In addition, vein recognition has advantages that other biometric identification technologies do not have, and has a wide range of application prospects, and has received a great deal of attention from scholars. Currently, vein feature extraction methods can be divided into handcrafted methods and deep learning methods.

A. Handcrafted Methods

Including geometric structure-based extraction [4], which mainly extracts the texture geometry of veins. Subspace-based methods [5], which specifically operate by mapping the high-dimensional features contained in the image to a low-dimensional space through projection to reduce the model complexity as much as possible without losing information, mainly include principal component analysis (PCA), linear discriminant analysis (LDA). Local invariant-based methods [6], such as Local Binary Pattern (LBP), Scale Invariant Feature Transformation (SIFT).

B. Deep Learning Methods

Deep learning based methods [7][8], the original vein image is trained using a convolutional neural network, the texture features of the veins are extracted and then the extracted features are sent to a classifier for classification, which is independent of the rotation, translation, scaling etc. of the image to obtain the desired results. Based on the results derived from other work. In general, Handcrafted Methods are more influenced by external factors. To reduce the influence of external factors, we choose a deep learning based approach to extract palm vein features, which has greater characterisation capability.

III. Method

The process of palm vein recognition is usually divided into three stages: image pre-processing, feature extraction and feature matching, where the segmentation of the region of interest (ROI) affects the overall classification effect. There are many methods for region of interest (ROI) segmentation, including methods based on the maximum inner tangent circle [9] and the centre of mass [10]. We next describe the methods used in this paper in detail.

A. Palm vein pre-processing

In the process of palm vein acquisition, we propose a new region of interest (ROI) segmentation technique in order to obtain a region of interest (ROI) containing more detailed information about the veins, because there are many differences in the position, angle and pose of the palm, as well as a lot of noise in the background information. The segmentation process consists of several steps: first, the palm vein image is binarised, and in order to obtain a more accurate binarised image, the palm vein image is Gaussian filtered before binarisation, and then the Gaussian filtered image is processed using Otsu's binarisation algorithm to obtain a binarised palm image, as shown in Fig. 1.

(a) (b) (c)

Figure 1. Image binarization process (a) Original palm image (b) Gaussian filtered image (c) Binary palm image

The coordinates of the centroid (x_0, y_0) are then calculated using Equation (1), where m and n denote the size of the image, x_i denotes the coordinates of row i, y_i denotes the coordinates of column j, and $f(i,j)$ denotes the grey scale value of row i and column j. We obtain the coordinates of the palm contour based on the binarised image, while marking the palm contour at the original image, as shown in Fig. 2.

$$x_0 = \frac{\sum_{i=1}^{m}\sum_{j=1}^{n} x_i f_{ij}}{\sum_{i=1}^{m}\sum_{j=1}^{n} f_{ij}}, \quad y_0 = \frac{\sum_{i=1}^{m}\sum_{j=1}^{n} y_j f_{ij}}{\sum_{i=1}^{m}\sum_{j=1}^{n} f_{ij}} \tag{1}$$

(a) (b)

Figure 2. Palm boundary detection (a) Centroid mark (b) Contour mark.

The next step is key point detection, where we use Equation (2) to calculate the Euclidean distance from the centroid to all points on the palm contour.

$$d = \sqrt{(x - x_0)^2 - (y - y_0)^2} \quad (2)$$

We remove the high frequency components from the function, thus removing the minima due to edge inhomogeneities, as shown in Fig. 3.

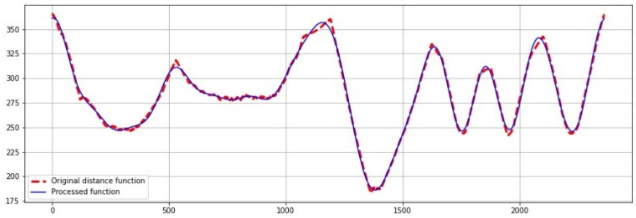


Figure 3. Euclidean distance from the centroid to the contour of the palm

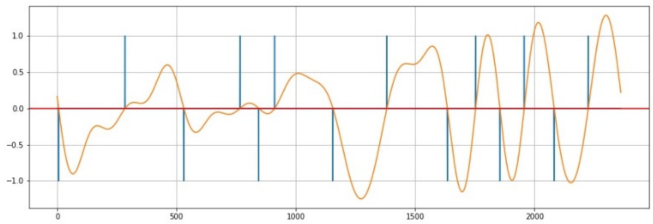


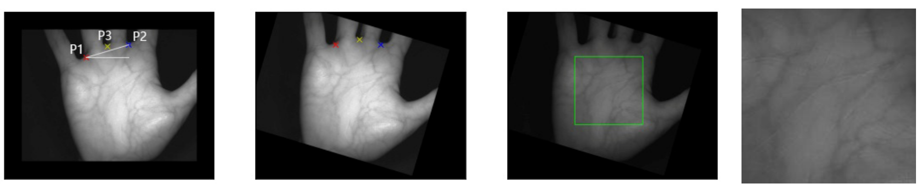
Figure 4. Differentiate to find the valley point from the local minimum

Next we differentiate the Euclidean distance function and then find the valleys from the local minima, as shown in Fig. 4. This in turn yields the coordinates P1, P2 and P3 for the valleys of the ring fingers and little fingers, the valley between the index and middle fingers, and the valleys of the middle and ring fingers, as shown in Fig. 5(a).

After obtaining the coordinates of P1, P2 and P3, we connect P1 and P2. The angle between the line $\overline{P1P2}$ and the horizontal line is the angle at which the image needs to be rotated, and we calculate the angle at which the image needs to be rotated based on equation (3) and the coordinates of P1 and P2, The rotated image is shown in Fig. 5(b).

$$\tan\theta = \frac{y_2 - y_1}{x_2 - x_1} \quad (3)$$

To obtain the ROI with more vein information, we use $\overline{P1P3}/2$ to the left of P1 as one endpoint of the ROI and $\overline{P2P3}/2$ to the right of P2 as the other endpoint of the ROI. However, the region of interest (ROI) containing more vein detail is in the centre of the palm, so it is not feasible to intercept the region of interest (ROI) from the key point, and for this consideration, we flatten the coordinates of P1 and P2 were simultaneously translated downwards by $\overline{P2P3}/2$, thus ensuring that more information about the veins of the palm was obtained, and then the distance between these two endpoints was used as the edge length of the ROI to intercept the region of interest (ROI) of the palm vein image, as shown in Fig. 5(c).The region of interest (ROI) is shown in Fig.5(d).



(a) (b) (c) (d)

Figure 5. ROI segmentation (a) refers to the valley mark (b) image rotation (c) ROI interception (d) Region of interest (ROI)

B. Improved MobileNet V2 model

The small sample problem can be largely solved using migration learning [11], which is based on the principle of first pre-training using one or more source domains and then applying the weights learned from the source domains to the target domain, migration learning aims to transfer the features learned in the source domain to the target domain. First, we sent the palm vein dataset to the modified MobileNet V2 model, and then loaded the palm vein pretraining except for the weights of the last layer into the palm vein training. Immediately afterwards, the palm vein dataset was trained using all layers of the modified MobileNet V2 network, with the palm vein data no longer trained from scratch and the model parameters shared between the palm print dataset and the palm vein dataset. The model parameters are then fine-tuned to fit the palm vein dataset and the desired classification results are obtained.

MobileNet V2 has the characteristics of lightweight [12]. The network uses deep separable convolutions. Deep separable convolutions use a smaller space cost and less time cost to achieve the same effect as a standard convolutional layer. Greatly reduces the amount of calculation and the size of the model. At the same time, the idea of ResNet [13] is adopted, and the inverted residual structure is added, which can effectively reduce the information lost after the ReLU activation function. The inverted residual structure first uses 1×1 convolution to increase the dimension, uses Depthwise convolution in high-dimensional space, and then uses 1×1 convolution to reduce dimensionality. 1×1 convolution and Depthwise convolution use ReLu6 nonlinearity Activation, 1×1 convolutional dimensionality reduction uses a linear activation function. As the depth of the network increases, the inverted residual structure can effectively solve the gradient disappearance or gradient explosion. At the same time, Batch Normalization (BN) [14] is added to the network, which can speed up the training speed and improve the generalization ability of the network. In order to increase the weight of important features, we have added Squeeze-and-Excitation Networks (SENet) [15] to MobileNet V2. The idea of SENet is to increase the weights of important features. As shown in Fig. 6.

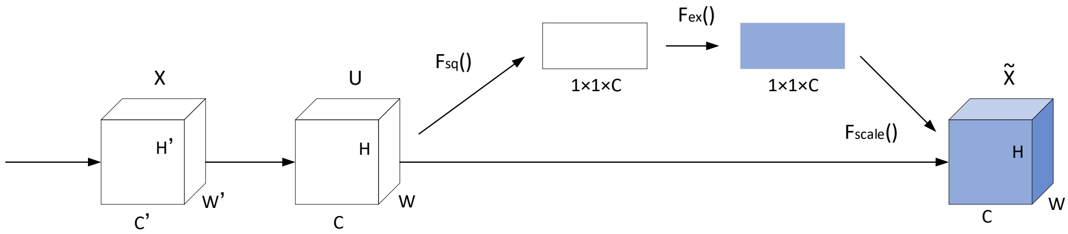


Figure 6. SE Module

We add SENet to the network after the Depthwise convolution in the inverted residuals. The residual structure with the SENet added is shown in Fig. 7. The network structure of our proposed SEM is shown in Table 1.

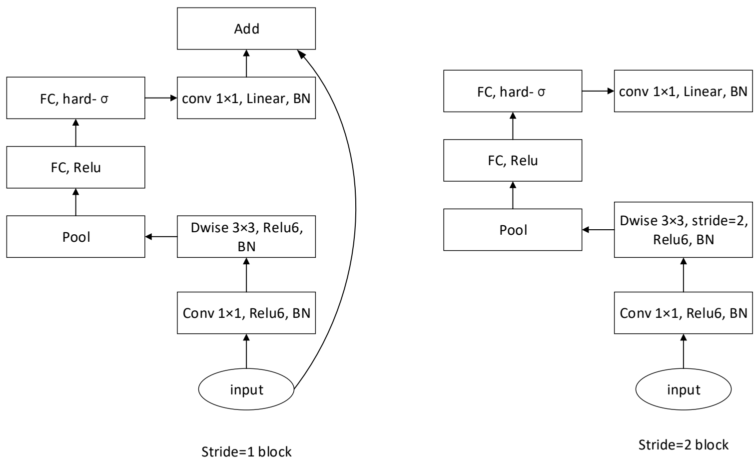


Figure 7. Add the Residual connection blocks of the SE module

Table 1. SEM network structure (t: expansion factor; c: number of output channels; n: number of module repetitions; s: first step of repeating n times).

Input	Operator	t	c	n	s
$224^2 \times 3$	conv2d	-	32	1	2
$112^2 \times 33$	bottleneck	1	16	1	1
$112^2 \times 16$	bottleneck	6	24	2	2
$56^2 \times 24$	bottleneck	6	32	3	2
$28^2 \times 32$	bottleneck,SE	6	64	4	2
$14^2 \times 64$	bottleneck,SE	6	96	3	1
$14^2 \times 96$	bottleneck,SE	6	160	3	2
$7^2 \times 160$	bottleneck,SE	6	320	1	1
$7^2 \times 320$	conv2d 1×1	-	1280	1	1
$7^2 \times 1280$	Avgpool 7×7	-	-	1	-
$1 \times 1 \times 1280$	conv2d 1×1	-	k	-	

MobileNet V2 uses a cross-entropy loss function as shown in Equation (4).

$$loss = -\sum_{c=1}^{M} y_c \log(p_c) \qquad (4)$$

where N is the number of categories, y_i is equal to 1 if the predicted category is the same as the sample marker, otherwise it is 0, and p_i is the probability that the sample is predicted to be of category i.

IV. Results

In this section, we compare the improved MobileNet V2 with other network models in terms of feature extraction and classification.

A. Palm vein dataset

The PolyU multispectral dataset was captured from 250 different people, these palm vein images were captured in two time periods, 6 images were captured for each individual palm in each time period, 12 images were captured for each individual palm, so the PolyU multispectral dataset contains 500 categories, 6000 palm vein images in total.

The Tongji University palm vein dataset contains 12,000 palm vein images. This dataset captures palm vein images from 300 different people. These images were captured in two sessions, with 10 palm vein images per hand per session, giving 40 palm vein images per person after the two sessions were completed.

The CASIA multispectral palm vein image dataset includes 7200 palm images collected from 100 different individuals. These palm images are 8-bit grayscale JPEG files. Thirty-six images were acquired for each hand over two time periods, with the two time periods separated by more than one month. Each image was captured simultaneously using six different electromagnetic spectra of the palm.

In pre-processing the images, we first fixed the aspect ratio of the images, then scaled the smallest edge to 256, and then cropped the images to a 224 x 224 size using centre cropping. We divided the dataset into a training set and a validation set in the ratio of 8:2, while to avoid overfitting the data, we used randomly rotated images as well as horizontally flipped images with a probability of 0.5.

B. Training details

In this this experiment, a cross-entropy loss function was used and we used Adam [16] to optimise the network parameters with a learning rate of 0.0005 at the very beginning, a batch size of 32 and Trained 100 epochs. This experiment was conducted using the Pytorch framework with training on a Tesla P100-SXM2 GPU.

C. Test results

The performance of the classifier is measured in terms of Accuracy Rate (ACC) and Equal Error Rate (EER). The accuracy rate (ACC) is the number of samples correctly classified divided by the number of all, as shown in Fig. 8. The loss of the validation set is shown in Fig. 9. EER is a biometric security system algorithm used to predetermine its error acceptance rate and its threshold of error rejection rate. When the False Rejection Rate (FRR) and the False Acceptance Rate (FAR) are equal, the values of FAR and FRR at this time are Equal Error Rates.A comparison of the accuracy and EER of our proposed method with other methods is shown in Table 2.

D.Experimental study of ablation of SEM

We tested four different network structures, i.e., MobileNet V2, SE + MobileNet V2, Transfer Learning + MobileNet V2, and SEM. From Table 3, we can see that our proposed SEM network model outperforms the other three, which is the reason why we chose the SEM network model.

Table 2. Comparison of Accuracy and EER with other methods.

Method	Accuracy(%)	EER(%)
LBP	0.8392	8.53
Gabor SIFT	0.9367	3.63
MobileNet V2	0. 9820	1.90
SEM	0.9935	1.49

Table 3. Results of ablation experiments.

Method	Accuracy
MobileNet V2	0.9820
SE + MobileNet V2	0.9873
Transfer Learning + MobileNet V2	0.9877
SEM	0.9935

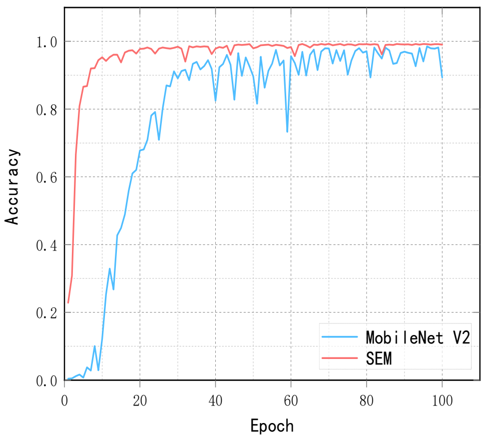


Figure 8. Comparison of trends in accuracy changes in the validation set

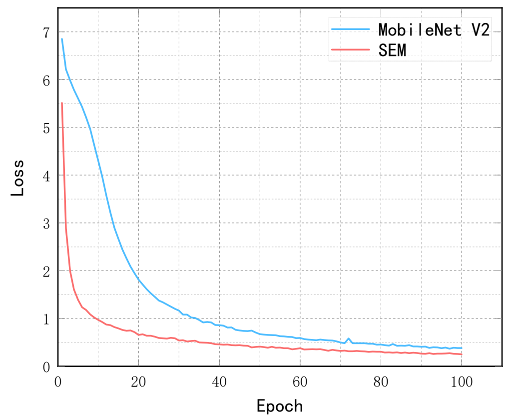


Figure 9. Comparison of loss variation trend of validation set

V. Conclusion

In this paper, we propose a SEM network model for palm vein recognition, and we redesign the coordinates of the intercepted ROI in order to obtain the region of interest (ROI) that contains more detailed information about the palm veins. To overcome problems such as overfitting caused by small datasets, we use transfer learning techniques and then use a fine-tuned MobileNet V2 model as the feature extraction method, and also add an attention mechanism to the MobileNet V2 model to increase the weight of important information. We conducted extensive experiments on the palm vein dataset and demonstrated that the proposed SEM model has better recognition results and significantly outperforms other methods in feature extraction and classification tasks.

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